

Smart Scaling: A Hybrid Deep-Learning Approach to Content-Aware Image Retargeting

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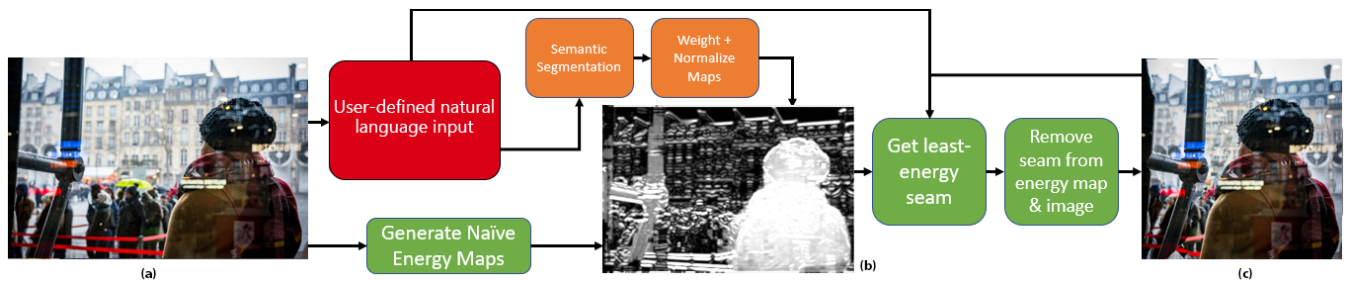


Figure 1: Introducing a combined pixel-gradient plus object segmentation energy model (b) into the seam carving pipeline results in a retargeted image (c) with significantly reduced distortion in the subjects. Photo source: Naftali Shavelson

KEYWORDS

computational photography, image processing

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1 INTRODUCTION

Images frequently need to be resized beyond simple cropping or stretching in order to preserve important content, where the choice of that content is often driven by artistic sensibilities or the use of the image (product advertising, texture creation, etc.). Existing automated resizing approaches, however, have failed to provide this art-director level control to select and preserve important aspects of the image [Rubinstein et al. 2010]. Automatic methods based on variations to the seam carving algorithm provide a powerful image-processing based approach for resizing which focus on automated saliency determination.

Seam carving is a method of content-aware image resizing in which an energy function determines which areas of an image to remove to preserve the primary subjects [Avidan and Shamir 2007].

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Recent developments to computer vision and natural language machine learning models allow for significant improvements to the seam carving pipeline for content-aware image retargeting.

We present a framework which augments an image's energy map with a confidence map generated by a segmentation model, allowing users to specify which subjects in an image to preserve using natural language prompts much like the role of an art director conversing with their artist. Using computer vision models to detect object positions also provides additional control over the composition of the retargeted image at various points in the image-processing pipeline. This object-based approach to energy map augmentation is incredibly flexible, because only minor adjustments to the processing of the energy maps can provide a significant degree of control over where seams are removed, how seam removal is prioritized in different sections of the image, and object and background separation and recomposition. This research explores how several different types of deep-learning models can be integrated into this pipeline in order to easily make these decisions, and provide different retargeting results on the same image based on user input and compositional considerations. Because this is a framework based on existing machine-learning models, this approach will benefit from advancements in the rapidly developing fields of computer vision and large language models and can be extended for further natural language directorial controls over images.

2 APPROACH

We propose three variations to the segmentation pipeline in this work, with varying degrees of manual control. The first method uses a deep-learning object detection model to automatically detect which objects are in the image. It then segments all object classes that were detected with a confidence above a given threshold. The

second method does the same, but limits the segmentation area to the bounding boxes of the detected classes. The third method uses GPT-4 to parse natural language prompts into segmentation classes and weights that are later used to adjust the weighting of each confidence map, allowing the user to organically describe their desired settings.

The use of natural language to direct the seam carving process demonstrates a significant improvement to the art-directability of content-aware retargeting, and yields strong results.

This approach uses the same base energy function as the original seam carving paper, calculating the difference in pixel values [Avidan and Shamir 2007]. The confidence maps are then used to artificially increase the energy of areas of the image that are part of a detected object with an augmented energy function:

$$e_{aug}(I) = w_s \cdot confidence(I) + \left| \frac{\partial}{\partial x} I \right| + \left| \frac{\partial}{\partial y} I \right| \quad (1)$$

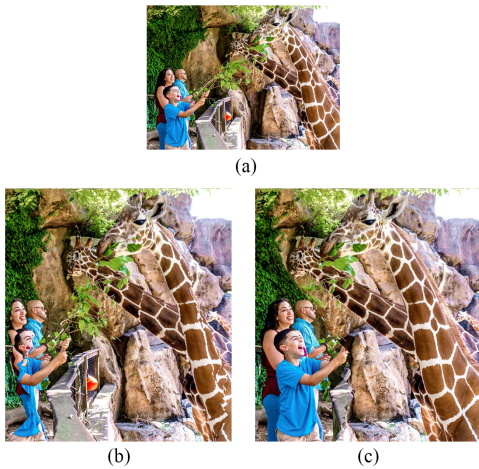


Figure 2: Original image (a), naive seam carving (b), hybrid retargeting preserving the family and the giraffes with natural language input (c). Photo source: Naftali Shavelson

One of the benefits to this pipeline is that any data from the object detection and segmentation models can be easily used to control where seams are prioritized. For example, object data can be used to control relative positioning of objects in the output image by removing seams in quantities proportional to the space between objects. This can be accomplished with dynamic programming by restricting the seam search areas and restricting the number of seams that the algorithm searches for in each area. In an image with width w and n detected objects, with each object O_n having a center point at O_{np} , the set of seams to be removed can be defined as $\{S\} = \bigcup_{i=1}^{n+1} S_i$, where each $\{S_n\}$ is a set of the $\frac{O_{n+1P} - O_{nP}}{w} k$ lowest energy seams in the space between each object. This will keep objects positioned relative to one another in the retargeted image, and could be beneficial in a larger image with fewer objects.

Additional considerations taking into account other compositional features, such as prioritizing different objects based on an object dominance hierarchy [Obrador et al. 2010], may be implemented as well. Calculating weights based on each object's size

and relative brightness could improve the performance of this retargeting approach in situations where there are too many seams to avoid objects completely.

Integrating this approach with a high-performance segmentation model provides the ability to carve seams from foreground objects separately from the background. More accurate segmentation allows for clean automated separation of layers in an image, each of which can be retargeted on its own. The retargeted layers can then be composited over each other again, with a generative model inpainting any gaps from object removal.



Figure 3: Retargeting the subjects and background separately. Original image, naive seam carving, hybrid retargeting. Photo source: Naftali Shavelson

This research also found that downscaling the original image before calculating the energy and upscaling the resulting energy map can significantly reduce processing time. Downscaling a 4k image to $\frac{1}{24}$ th the original image size resulted in a 90% reduction in seam carving time with no visual impact to the output image.

3 CONCLUSION AND FUTURE WORK

Retargeting images with this method can result in improved output compared to the naive seam carving approach, and provides natural-language art-directability. Improvements are especially noticeable in images with low-contrast subjects or busy backgrounds. The flexibility of the pipeline allows for easy implementation of improved computer vision models as they are developed, and provides an additional framework for art directability that many other content-aware retargeting models lack.

Further work can be done to improve this framework, such as introducing forward-energy [Rubinstein et al. 2008] or more advanced models to better preserve background elements. Cases in which a subject takes up the entire width of the image still present problems, and can be improved upon in further research.

This pipeline can benefit from new developments in computer vision, and as new models are developed with improved performance, the performance of this pipeline will improve as well.

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